College Search & Intelligence Platform

AI-Powered Academic Assistant

Transforming how students and teachers interact with educational materials through cutting-edge Generative AI and Vector Search.

The Problem: Information Overload

- **Scattered Resources**: College students receive hundreds of PDFs, syllabi, and notes across various platforms.
- **Wasted Time**: Finding a specific concept (e.g., "What is normalization in DBMS?") requires manually skimming dozens of documents.
- **Lack of Interaction**: Static PDFs don't answer questions or test knowledge.

The Result: Students spend more time *searching* for information than actually *studying* it.

Our Solution & Workflow

A dual-role, AI-driven platform that makes documents "smart".

1. **Teachers/Admins** upload course materials (PDFs).
2. **AI Microservice** extracts text, breaks it into logical chunks, and generates **Vector Embeddings** (mathematical representations of meaning).
3. **Students** ask questions in natural language.
4. **Semantic Search** finds the exact paragraphs needed.
5. **Generative AI (LLaMA 3)** reads those paragraphs and provides a direct, synthesized answer.

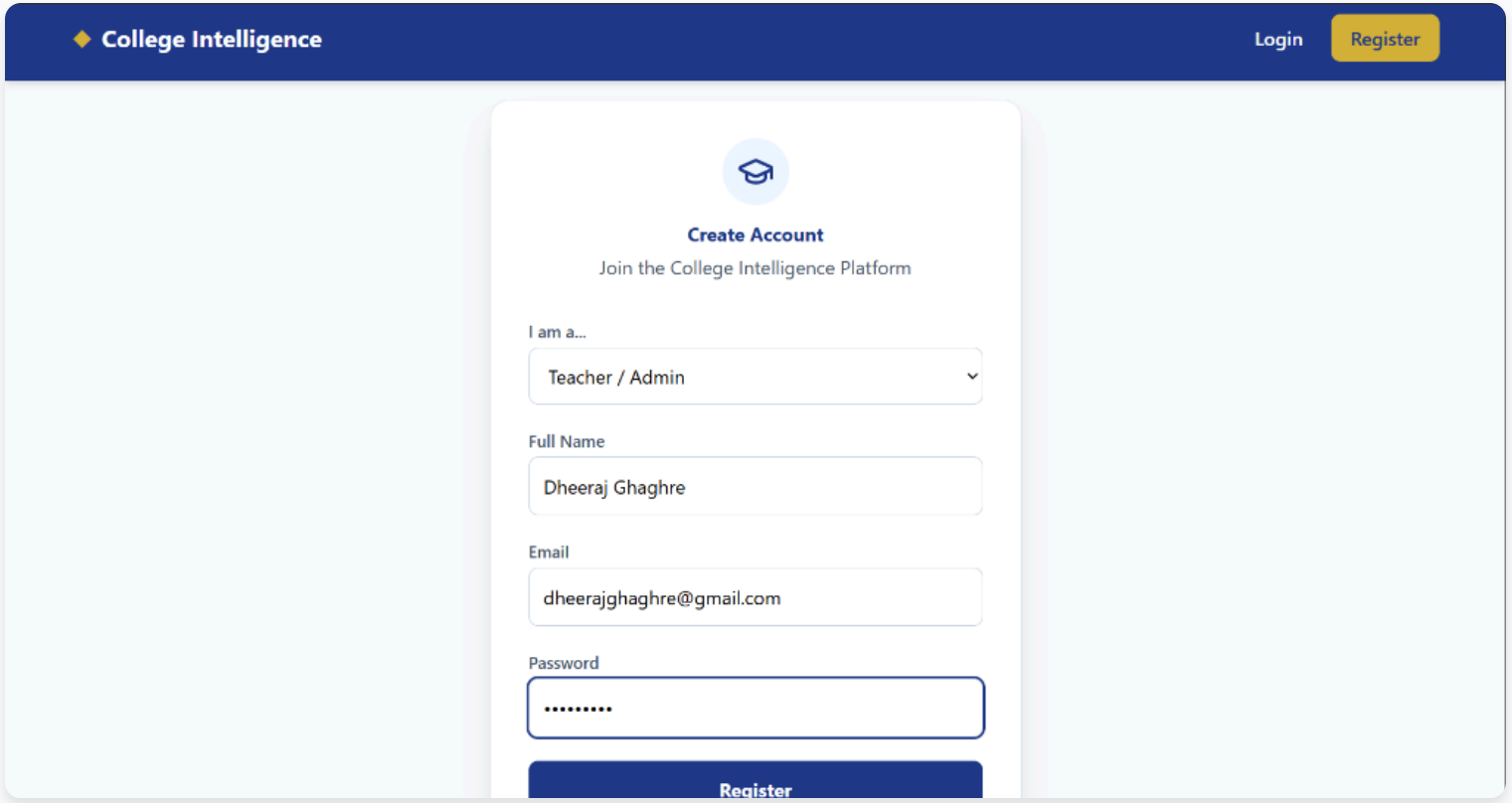
Technology Stack

We built a modern, scalable, and modular architecture:

- **Frontend (UI)**: React, Vite, Tailwind CSS, TypeScript
- **Backend API**: Node.js, Express.js
- **Database**: PostgreSQL (managed via Prisma ORM)
- **AI Microservice**: Python, FastAPI
- **AI Models**:
  - `sentence-transformers` (`all-MiniLM-L6-v2`) for semantic chunking & vectors.
  - **Meta Llama 3 8B** (via Groq) for high-speed generative answers.

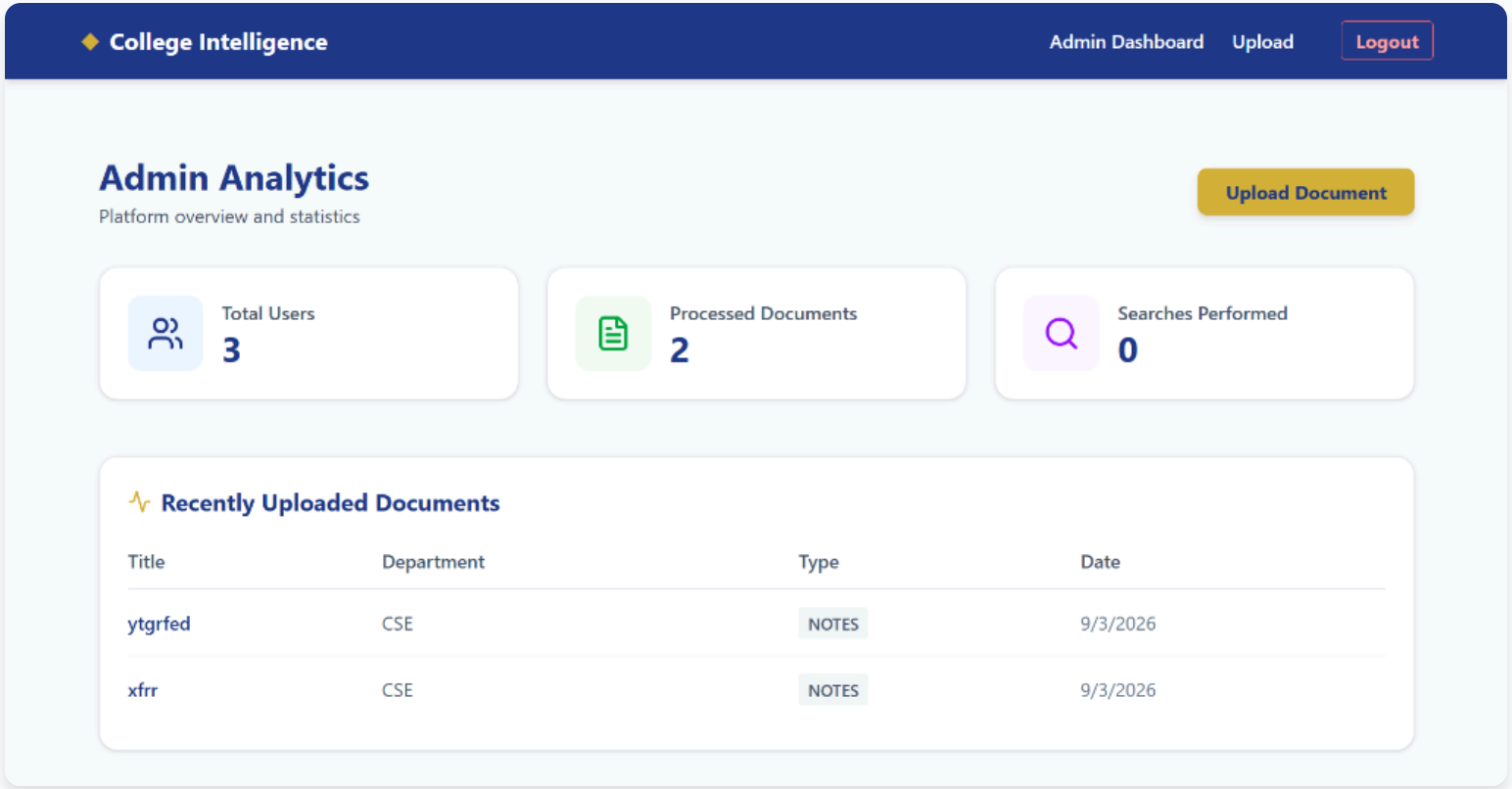
1. Secure Access & Role Management

The platform utilizes JWT-based authentication with strict Role-Based Access Control (RBAC). Users register as either a **Student** or a **Teacher/Admin**, which dynamically routes them to customized experiences.



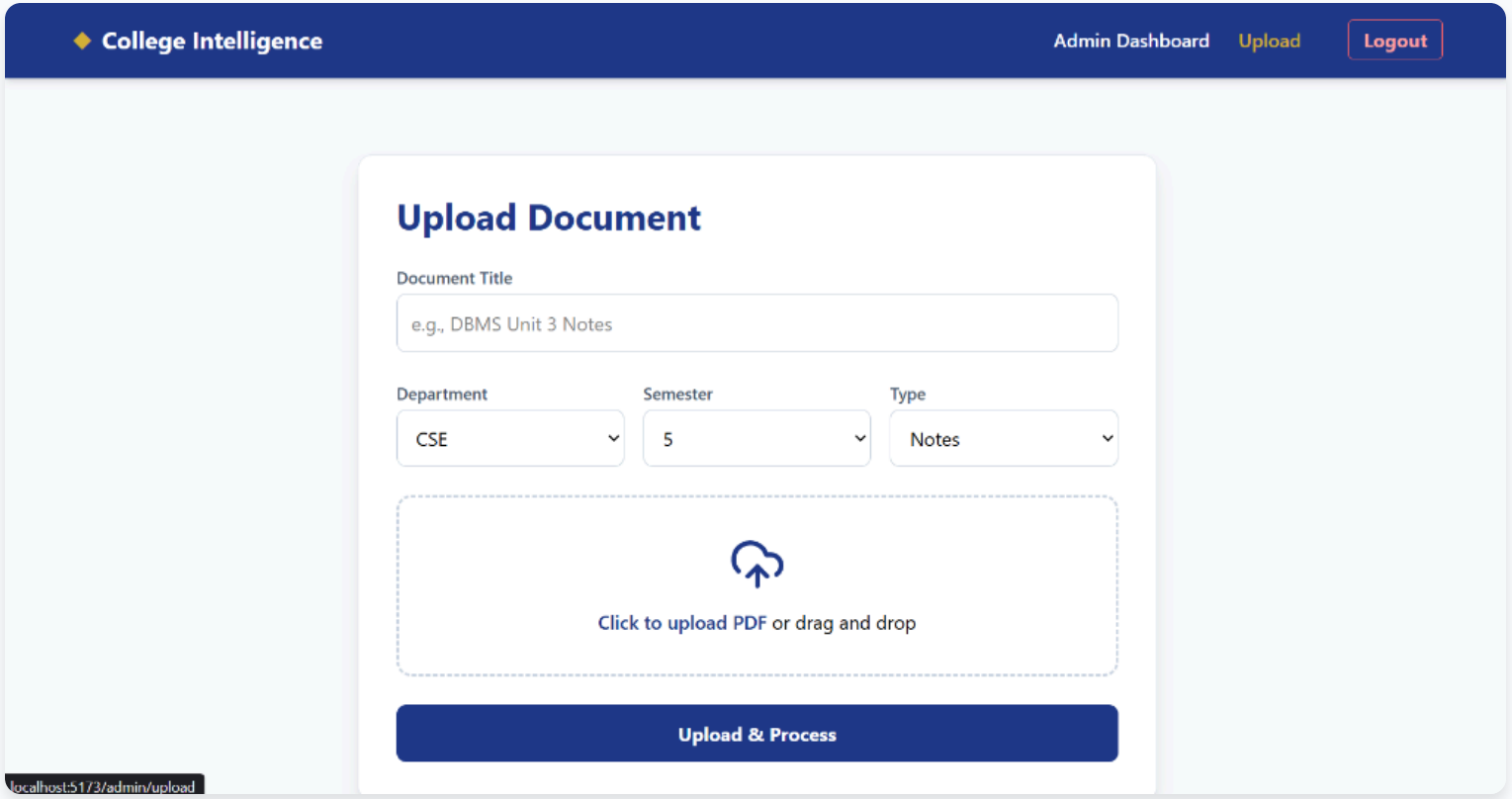
2. The Teacher Experience

Teachers have access to the **Admin Analytics Dashboard**. They can monitor total platform usage, track processed documents, and manage the knowledge base.



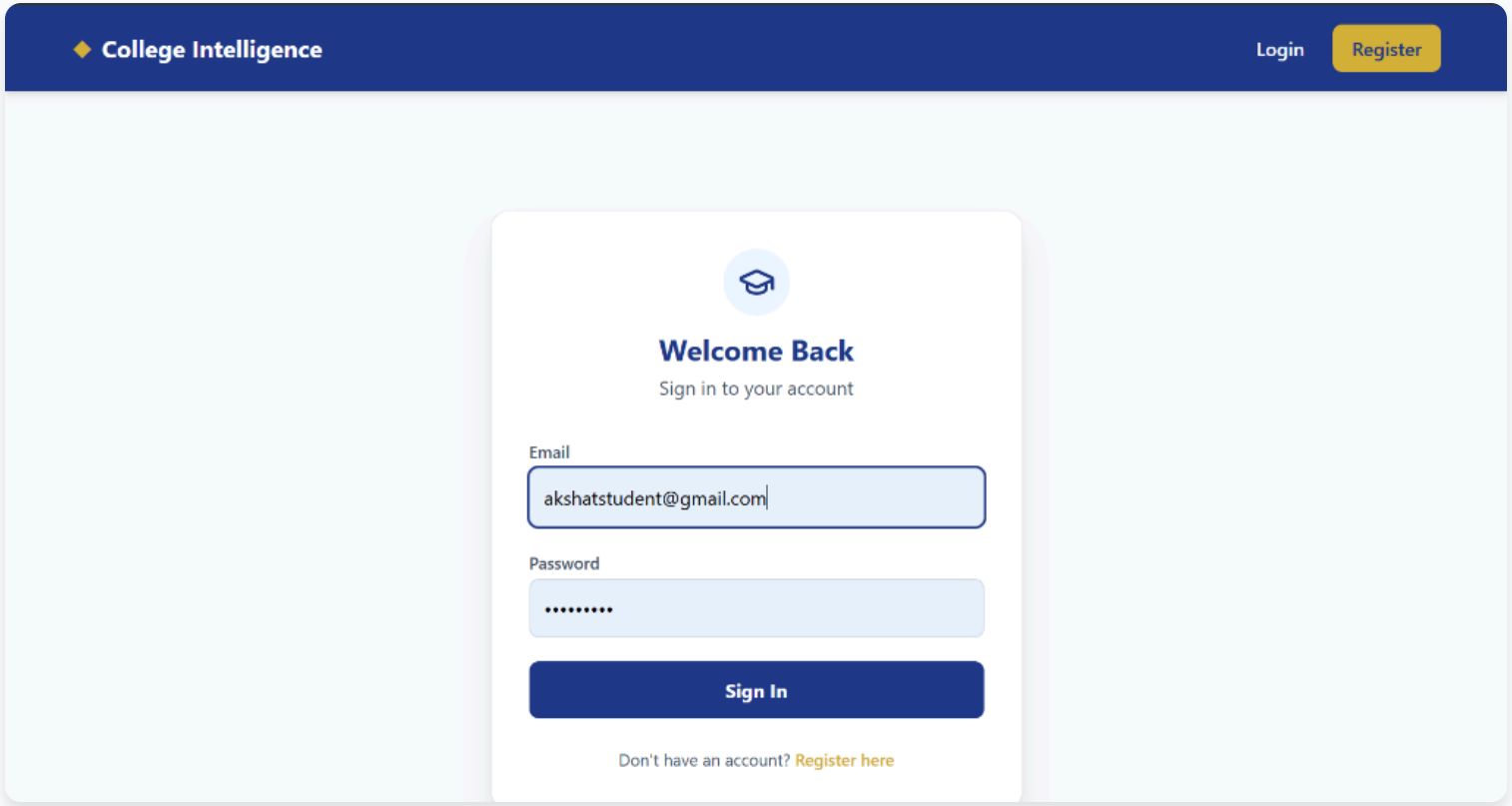
3. Document Processing Pipeline

When a Teacher uploads a document, the system automatically deduplicates it (using SHA-256 hashing) and forwards it to the Python AI service for text extraction and vector embedding.



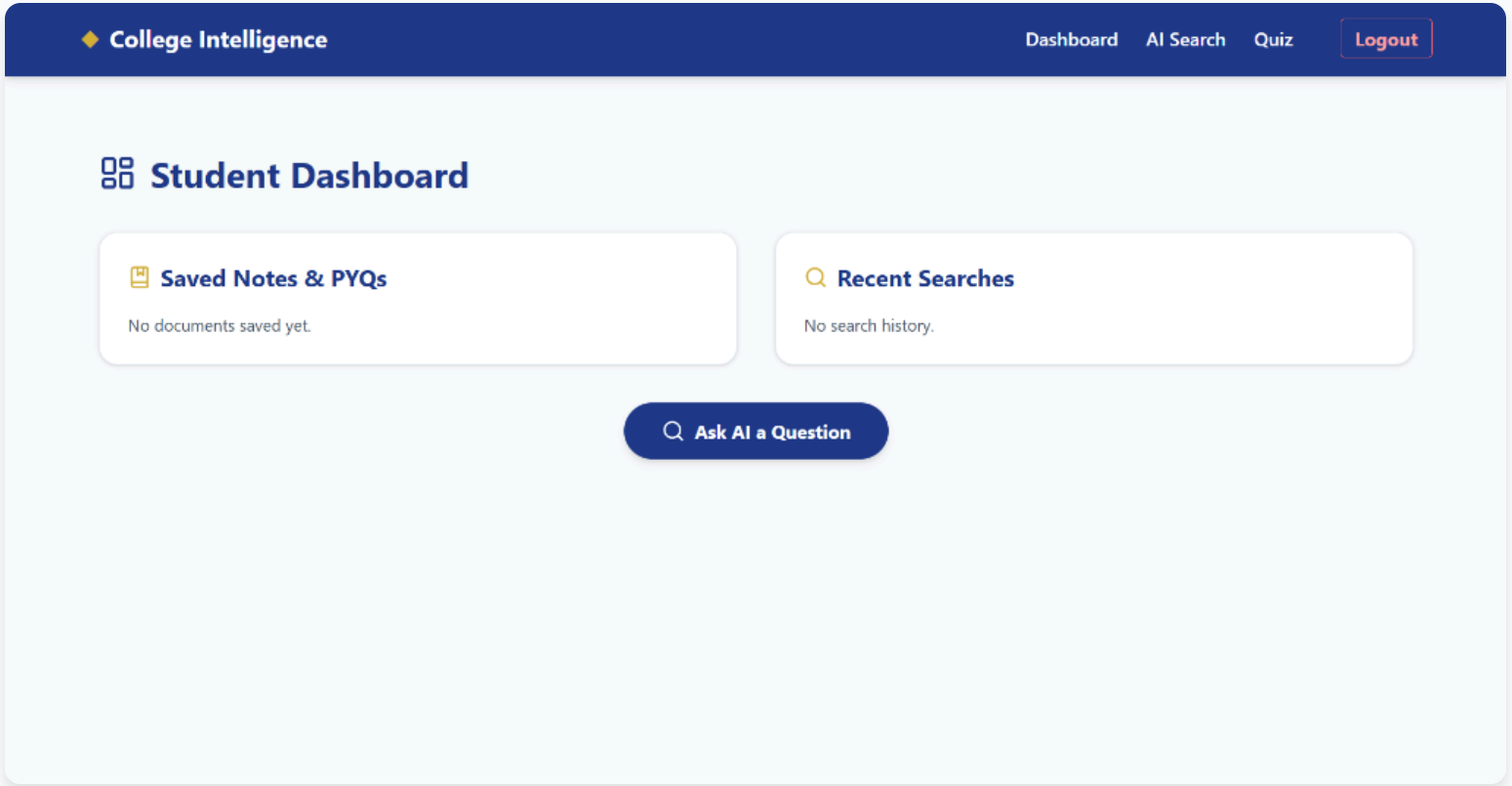
4. The Student Experience

Students log in to a distraction-free environment tailored for studying.



5. Interactive Learning

The Student Dashboard provides instant access to saved notes, recent search history, and the core feature: **Asking the AI a question**. The platform acts as a 24/7 personalized tutor based *strictly* on the college's uploaded curriculum.



Conclusion & Future Scope

Impact:

- Significantly reduces study friction and search time.
- Ensures AI answers are grounded in official course materials (No hallucinations).

Future Scope:

- Automated Quiz & Flashcard generation from documents.
- Peer-to-peer note sharing and collaborative workspaces.
- Support for images and handwritten note parsing via OCR.

Thank You!